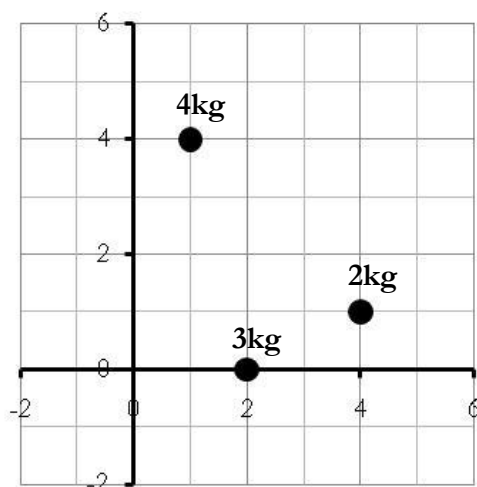
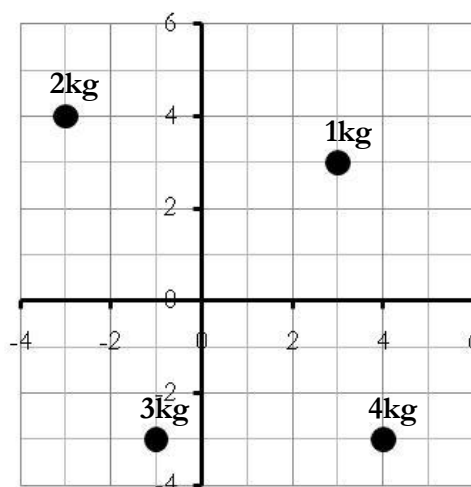


1. Find the coordinates of the centre of mass of each of the following systems of particles.

(a)

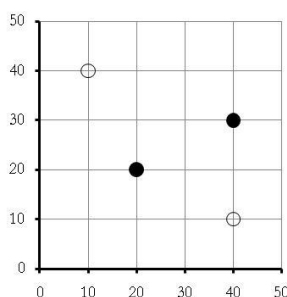


(b)



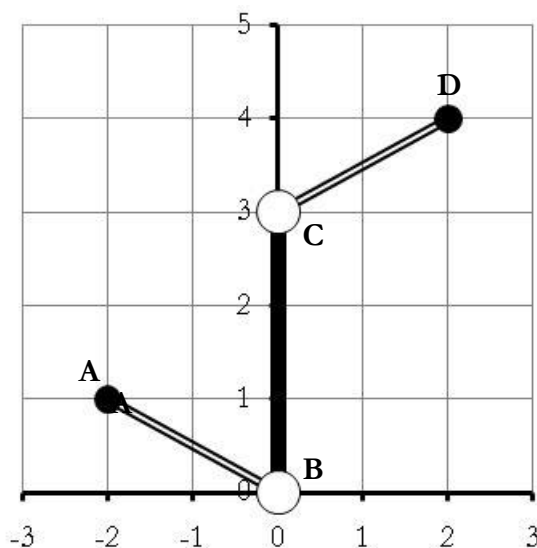
2. Three particles of mass 2 kg, 1 kg and 3 kg are situated $(4, 3)$, $(1, 0)$ and (a, b) respectively. If the centre of mass of the system lies at $(0, 2)$, find the values of a and b .
3. The rectangle ABCD has $AB = 4$ cm and $AD = 2$ cm. Particles of mass 3 kg, 5 kg, 1 kg and 7 kg are placed at the points A, B, C and D respectively.
- Find the distance of the centre of mass of the system from each of the lines AB and AD.
 - The rectangle is then suspended from the point A. Find the angle that AB makes with the vertical.
[Hint: do **not** attempt to draw a rotated diagram. Simply draw in the “new” vertical and annotate it as such.]
 - An *additional* mass of 4 kg is added at D. The object is once again suspended from A. Find the angle that AB makes with the vertical.
[Hint: it is **not** necessary to start again.]
 - The mass added in (c) remains in place, but the mass of mass of 5 kg is removed from B. The object is once again suspended from A. Find the angle that AB makes with the vertical.
[Hint: it is **not** necessary to start again.]

4. Two full mugs (shown by coloured dots) and two empty mugs (blank dots) are placed on a large square tray of side length 50 cm. Each mug has mass 1 kg and the liquid inside each of the full mugs has mass 0.5 kg.



A waiter carries the tray by balancing it on one hand.

- If the tray is plastic (and can be modelled as light), determine where he must hold the tray so that it will balance.
 - If the tray is metal and uniform, and has mass 3 kg, determine where he must hold the tray so that it will balance.
5. Three particles of mass 1 kg, 2 kg and m kg are situated at $(5, 2)$, $(1, 5)$ and $(1, -2)$ respectively. If the centre of mass of the system lies at $(2, \bar{y})$, find the values of m and $\bar{y}$.
6. A model of a molecule hangs in a chemist's laboratory. It is composed of two heavier 'atoms' (at B and C) each of mass 2 kg and two lighter 'atoms' (at A and D) each of mass 1 kg. The rods connecting A to B and C to D are light, the rod connecting B to C is uniform and has mass 2 kg.



- Find the position of the centre of gravity of the model.
- If the model is suspended from D, find the angle that CD makes with the vertical.
- If the model is suspended from A, find the angle that AB makes with the vertical.